



ELECTRONICS AND CONTROLS

WEEK 1 - SUBMISSION

By:- Arka Dasgupta (2025ACPS1266G)

[21 June 2026]

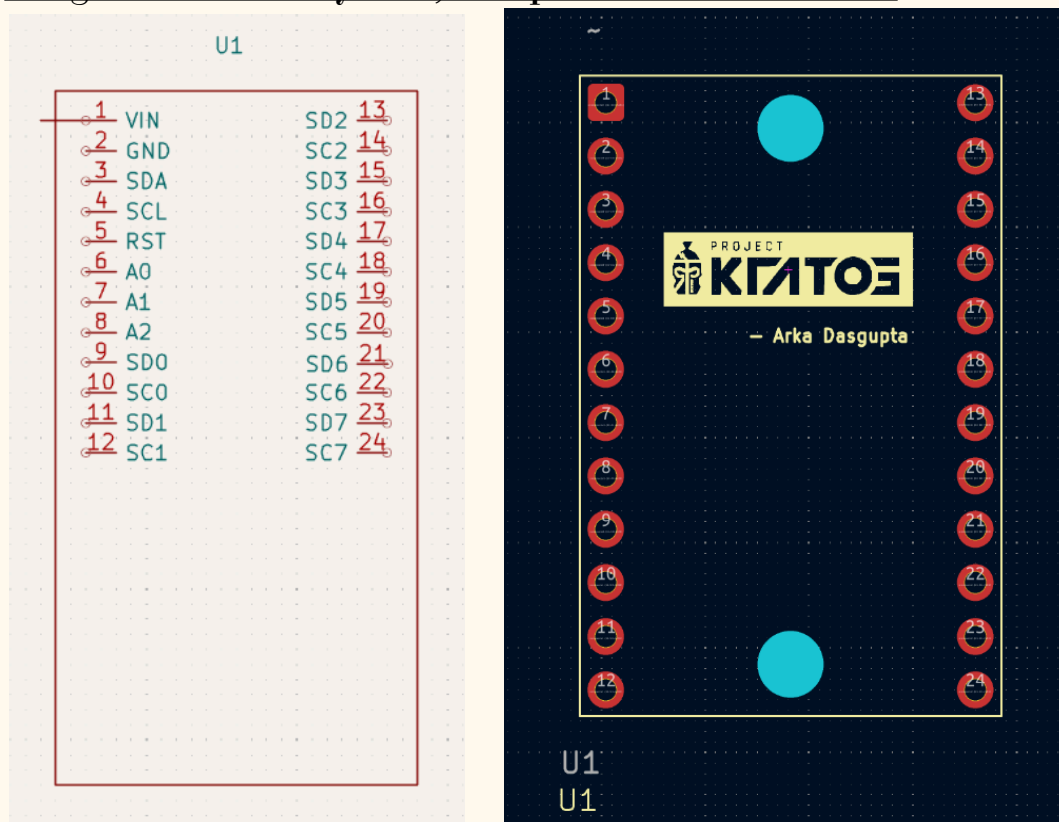
~ ASSIGNMENT 1 ~

Assignment Name:- Making a Custom Part (TCA9548A I2C Multiplexer)

Link to Assignment (GitHub):-

https://github.com/arkadasgupta07/ProjectKratos/tree/main/Hardware/I2C_Multiplexer

Images of Custom Symbol, Footprint and 3D Render:-





Hardware:-

1. Created a custom 24-pin schematic symbol for the TCA9548A I2C Multiplexer.
2. Designed a custom 24-pin through-hole footprint with custom Project Kratos text and logo on the silkscreen.

No. of Layers:- 2 Copper Layers

Copper Layer Thickness:- 0.035mm (35 μm) or 1 oz/ft²

Part Selection Justification:-

1. **Connectors:-**
 - **Control Pins (Pins 1–8):** 2.54mm pin headers for power (3.3 V/GND), main I2C signals (SDA, SCL), reset pin, and address selection pins (A0–A2).
 - **Output Channels (Pins 9–24):** 2.54mm pin headers that split the main I2C line into 8 separate sub-channels (SD0/SC0 to SD7/SC7).
2. **Buck converters/resistors/other modules:-**
 - **TCA9548A Multiplexer Chip:** Used to switch between 8 different I2C channels so multiple sensors with the same address can talk without conflicts.

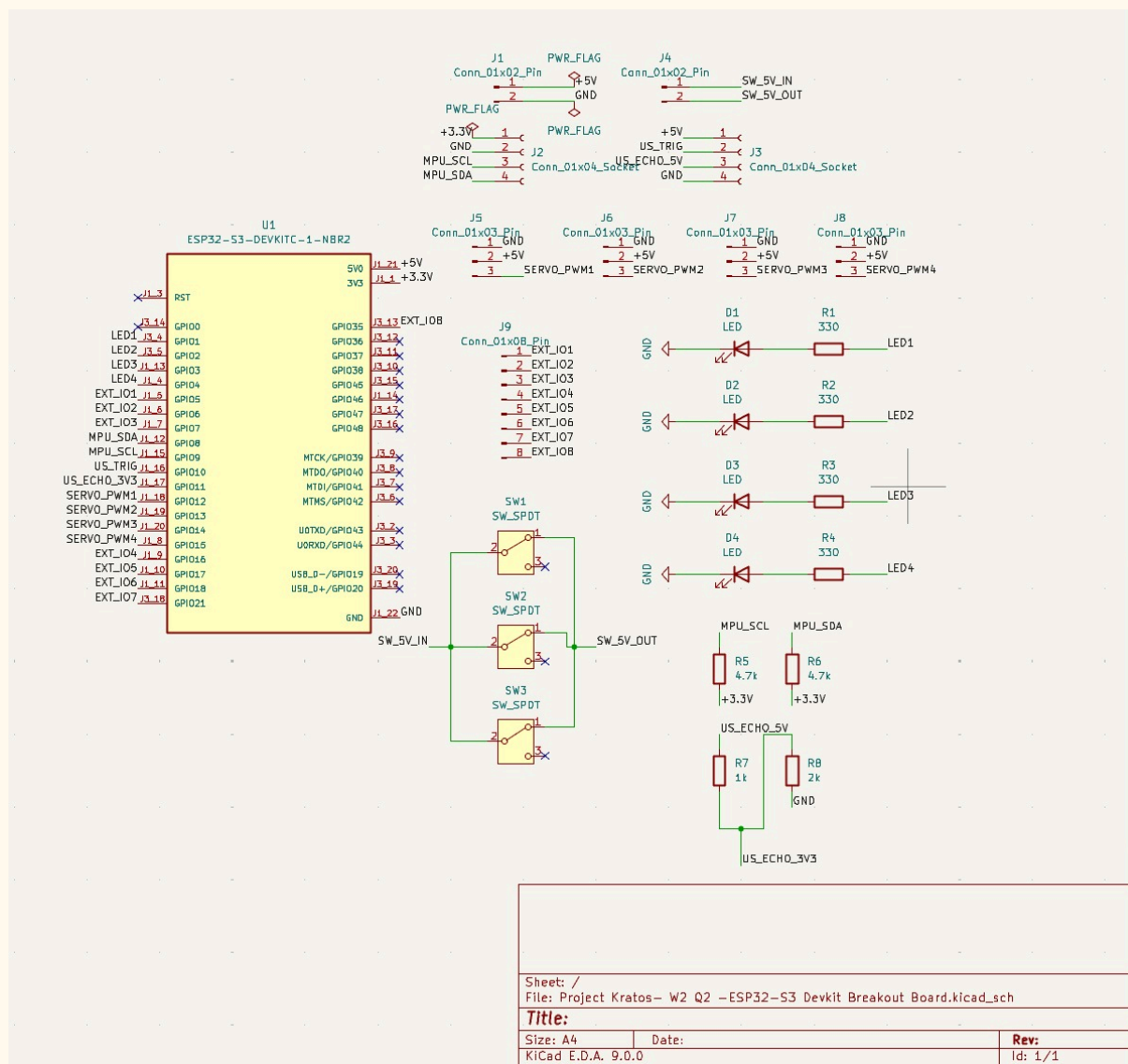
~ ASSIGNMENT 2 ~

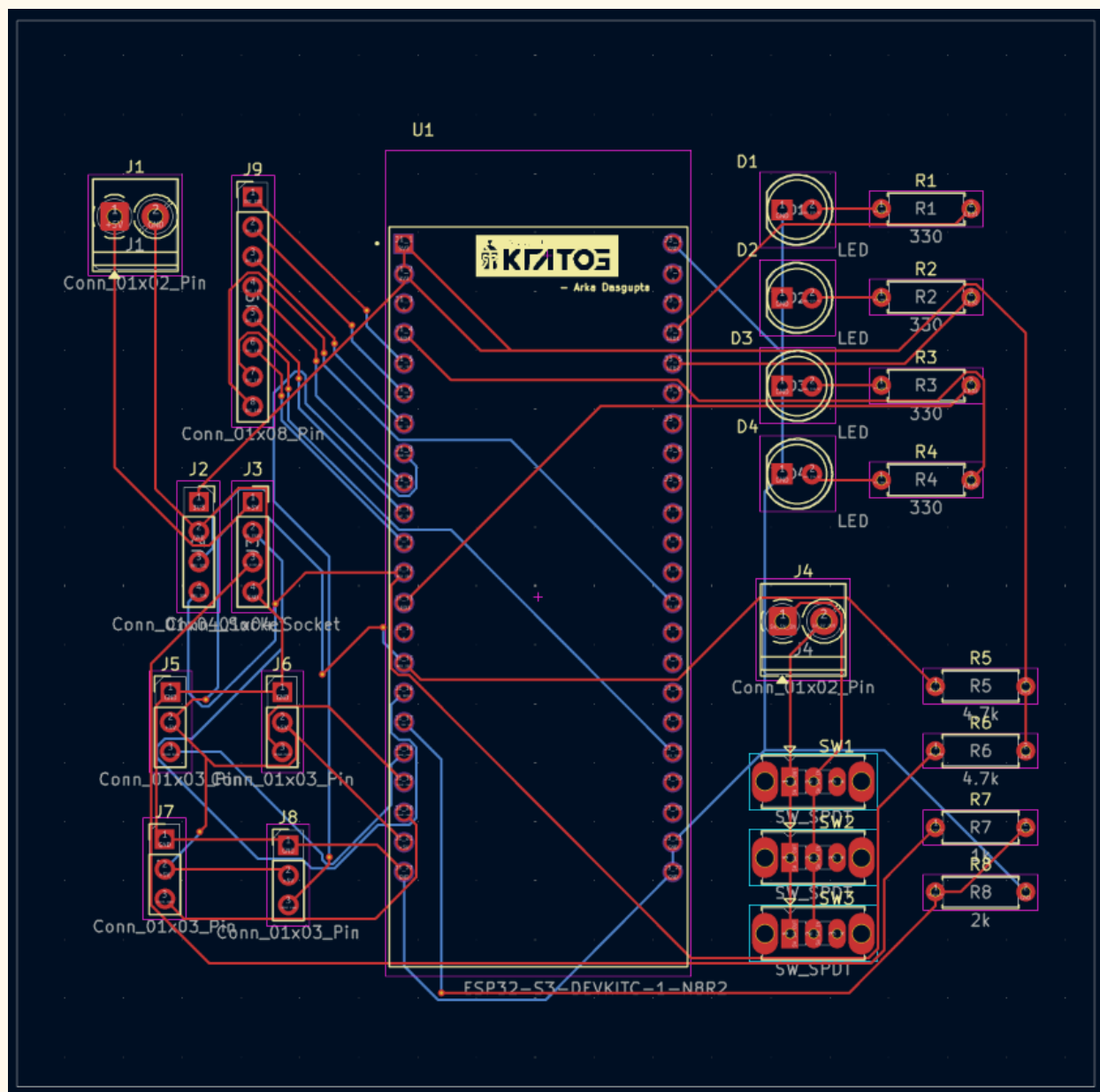
Assignment Name:- ESP32-S3 DevKit Breakout Board

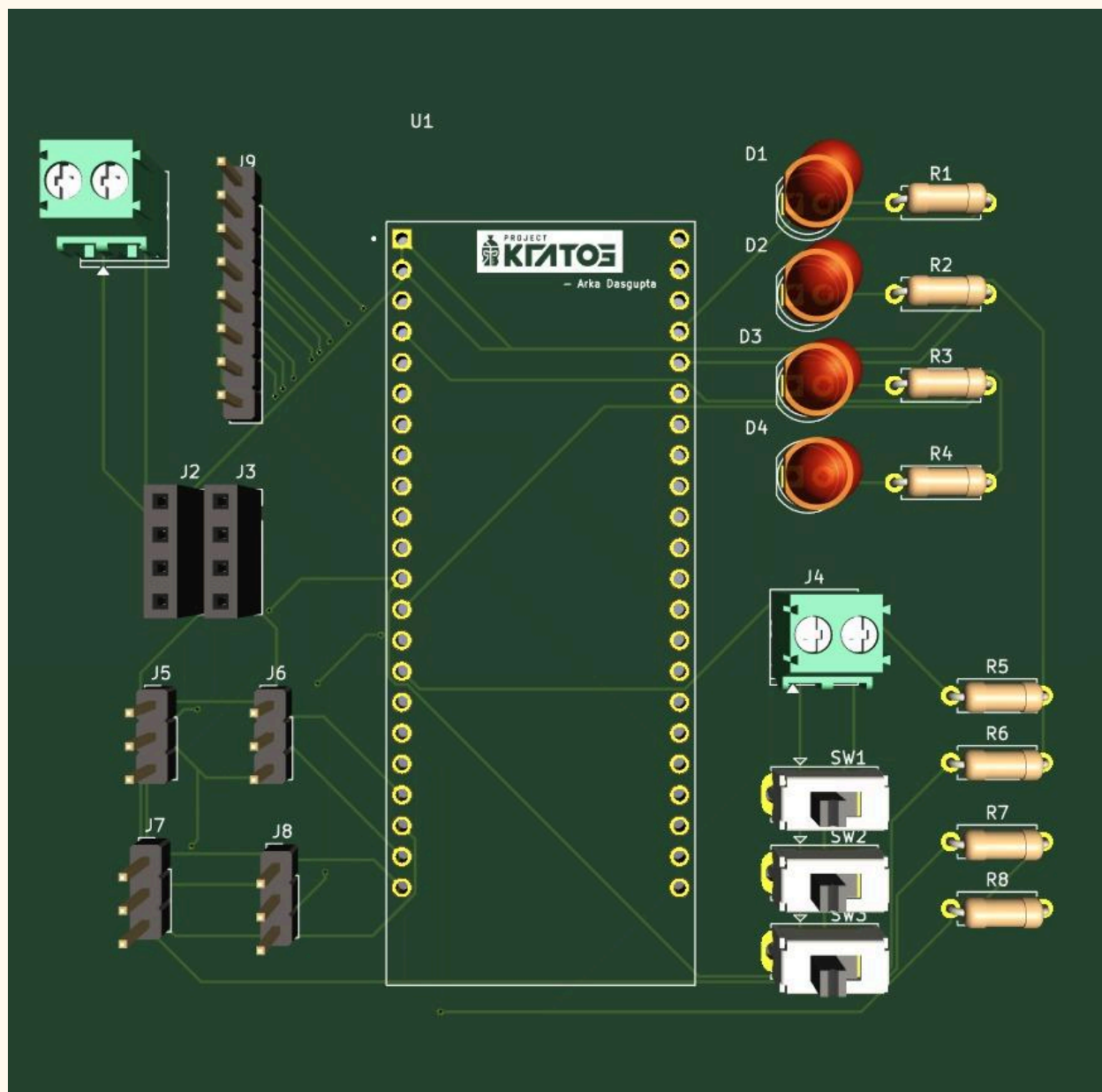
Link to Assignment (GitHub):-

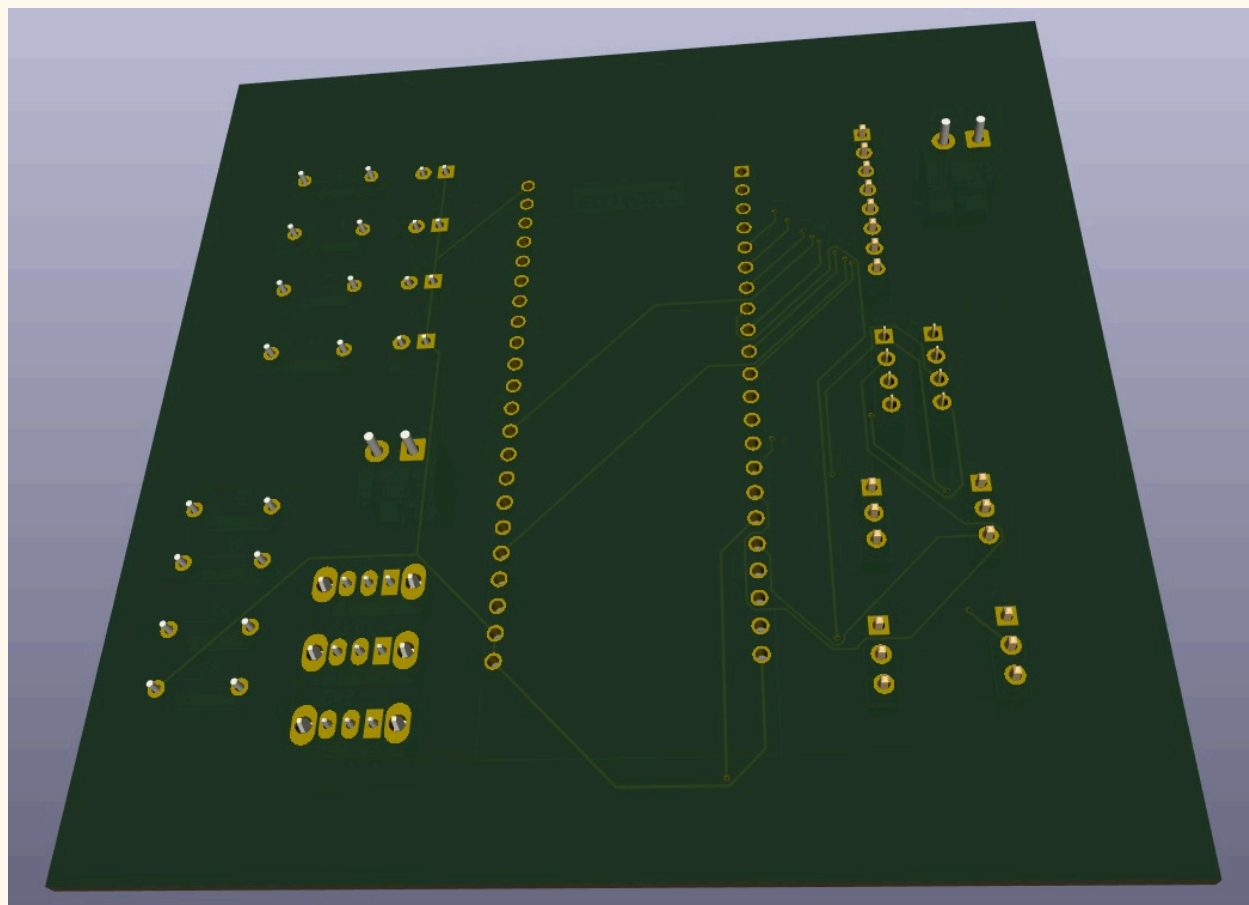
https://github.com/arkadasgupta07/ProjectKratos/tree/main/Hardware/ESP32S3_Breakout_Board

Images of Custom Symbol, Footprint and 3D Render:-









Hardware:-

1. ESP32-S3-WROOM-1 module for Wi-Fi and Bluetooth processing.
2. USB Type-C port for power and uploading code directly from a laptop.
3. 3.3V LDO voltage regulator chip.
4. Reset/Boot buttons.
5. Status LEDs with resistors.

No. of Layers:- 2 Copper Layers

Copper Layer Thickness:- 0.035mm (35 μm) or 1 oz/ft²

Part Selection Justification:-

1. **Connectors:-**
 - **USB Type-C Connector:** Added for modern, convenient power supply and fast code flashing.
 - **Breakout Pin Headers:** 2.54mm male headers on both sides to expose power and GPIO pins for easy breadboard wiring.
2. **Buck converters/resistors/other modules:-**
 - **3.3V LDO Voltage Regulator:** Safely converts 5V power coming from the USB cable down to 3.3V needed by the ESP32-S3.
 - **Decoupling Capacitors (100nF, 10 μF):** Filters out noise in the power lines to prevent the microchip from crashing.
 - **EN & BOOT Circuits:** Resistors and small push-buttons connected to easily reset the board or put it into download mode.

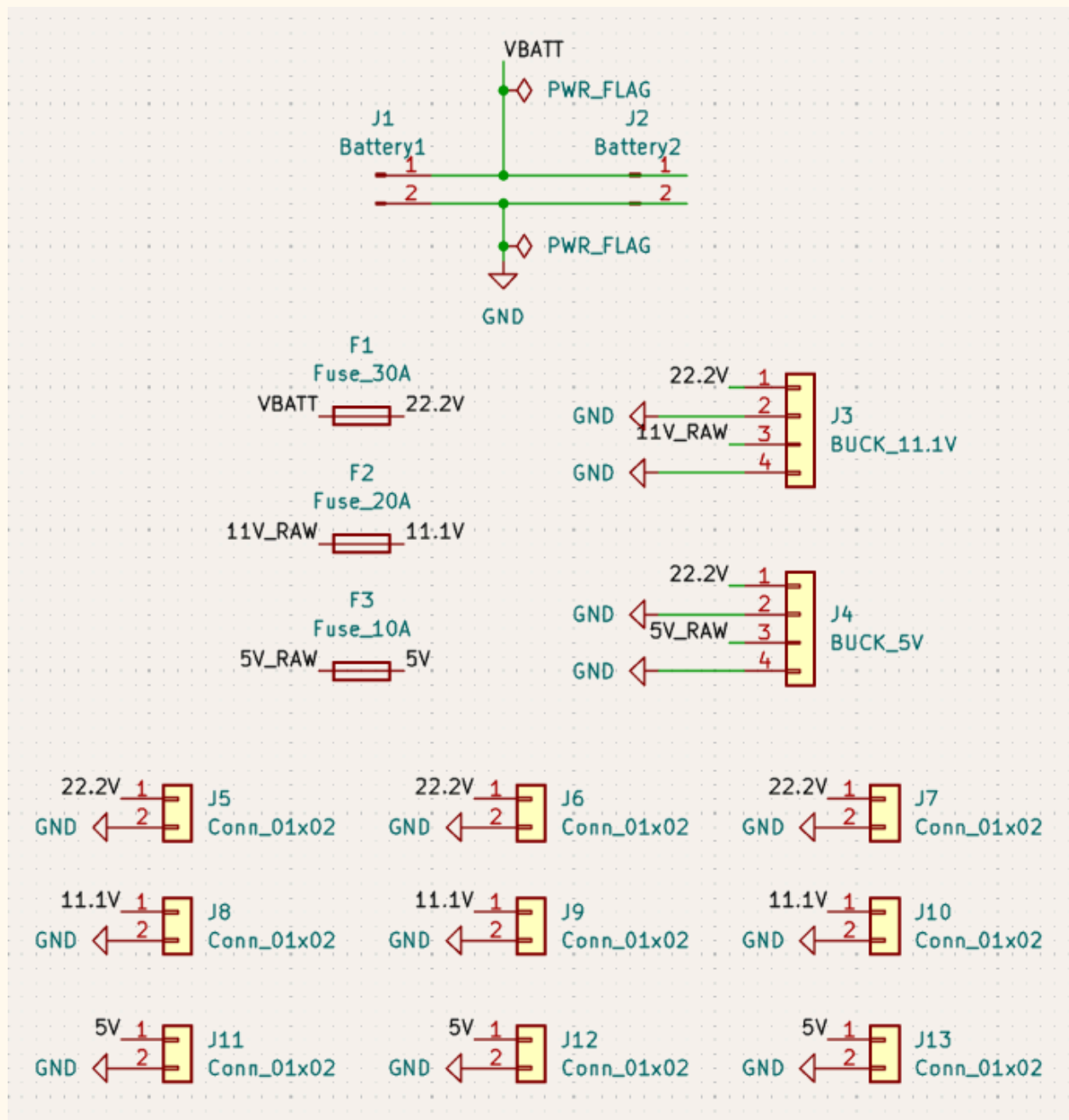
~ ASSIGNMENT 3 ~

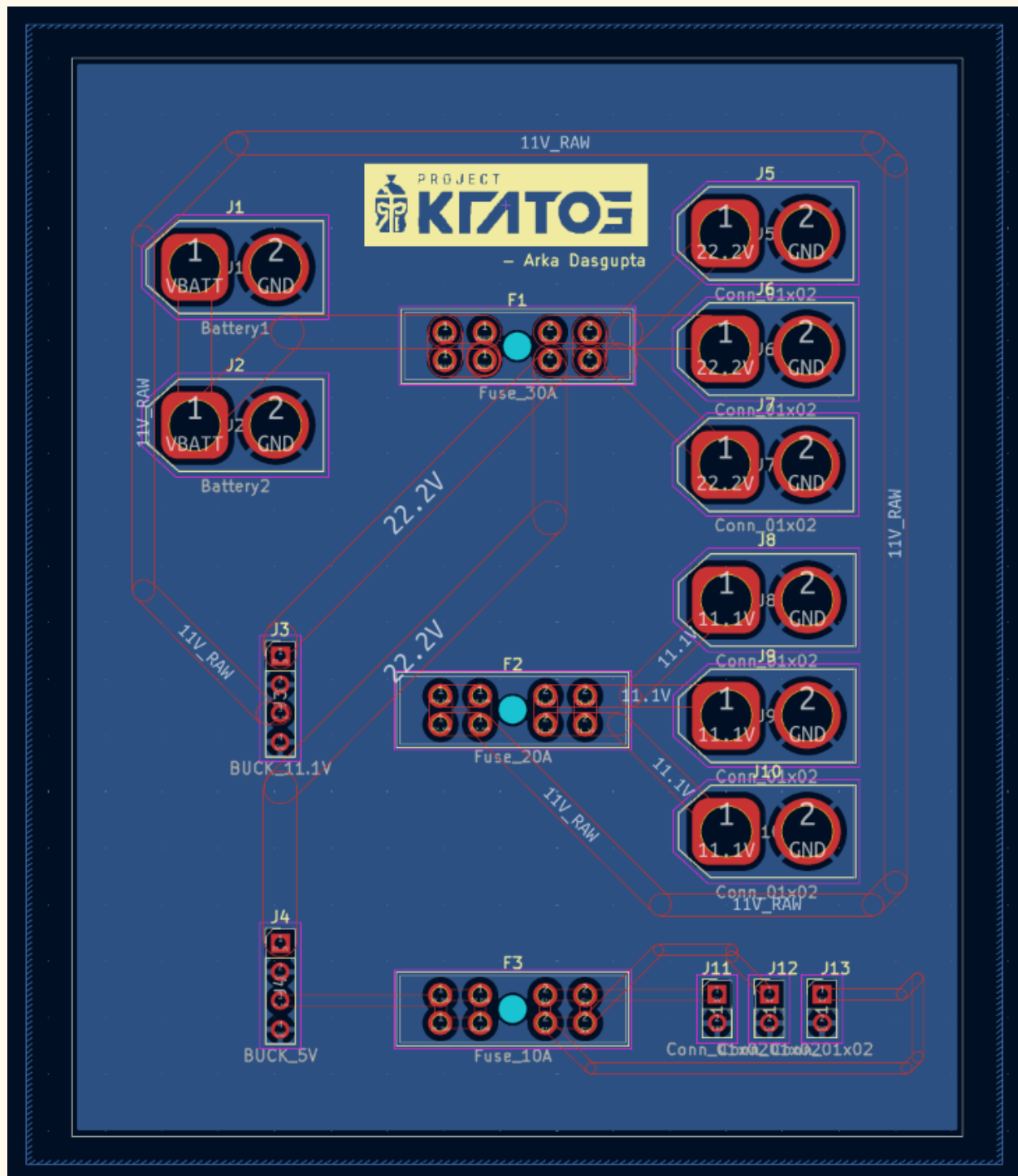
Assignment Name:- Power Distribution Board

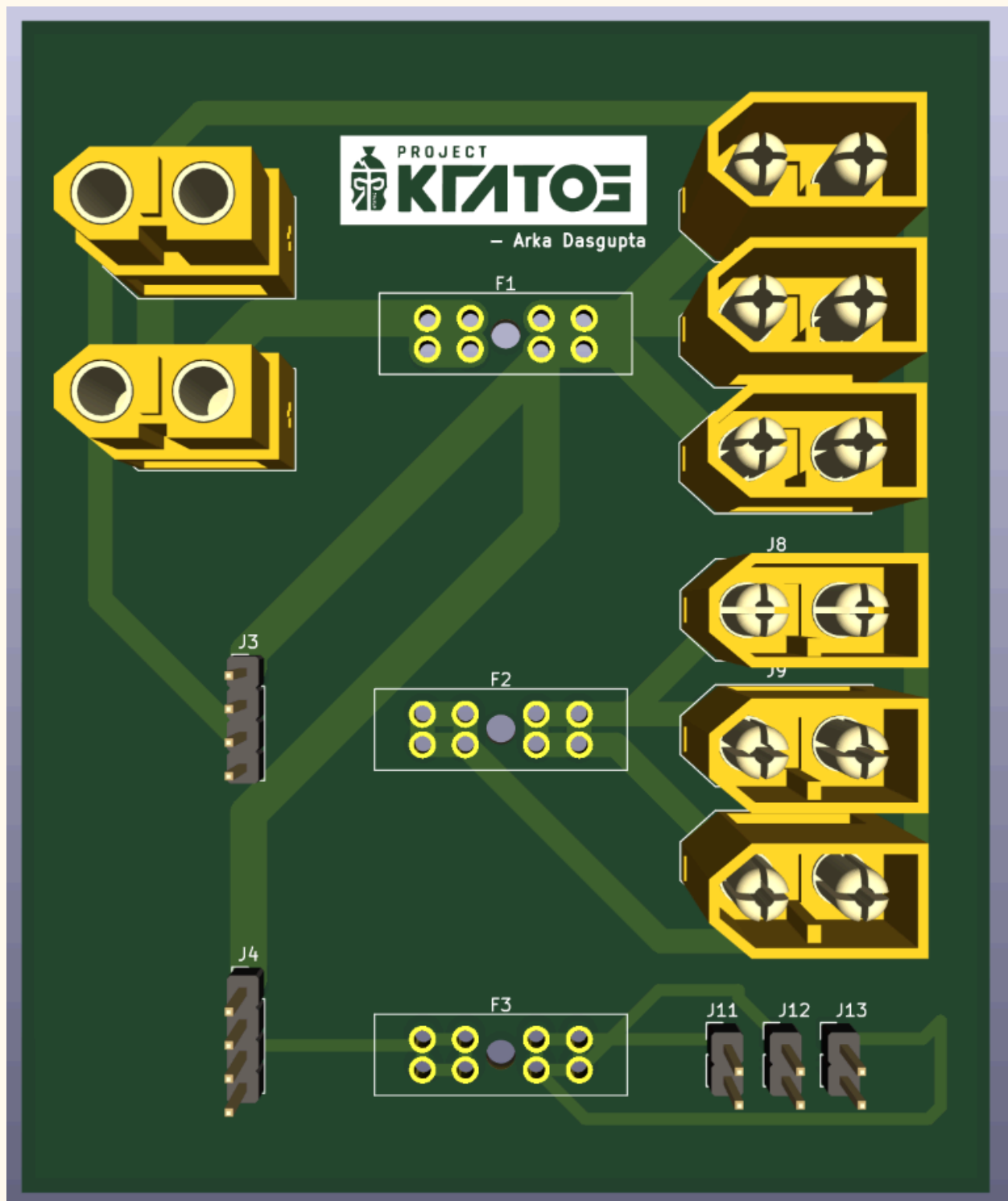
Link to Assignment (GitHub):-

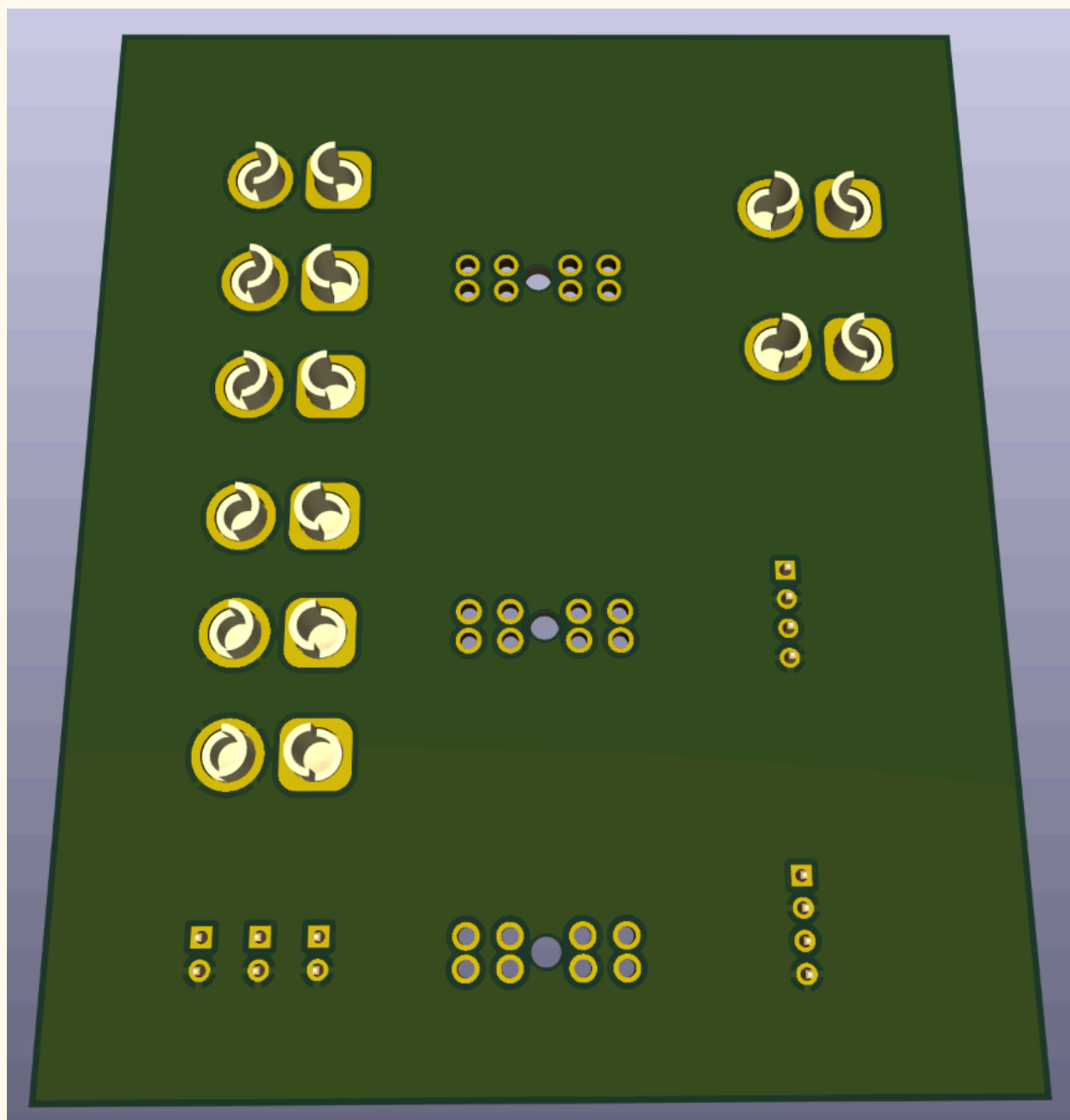
https://github.com/arkadasgupta07/ProjectKratos/tree/main/Hardware/Power_Distribution_Board

Images of Custom Symbol, Footprint and 3D Render:-









Hardware:-

1. High-current XT60 connectors for main power input and output distribution.
2. Blade Fuse / Inline Fuse holders (F1, F2, F3) for overcurrent protection across individual power rails.
3. Auxiliary 2.54mm pin headers (J3, J4, J8–J13) for logic signals, telemetry, or low-power taps.
4. Wide copper trace routing designed for minimal resistance and higher current capacity.
5. Custom branding logo on the front silkscreen layer (F.Silkscreen).

No. of Layers:- 2 Copper Layers

Copper Layer Thickness:- 0.035mm (35 μm) or 1 oz/ft²

Part Selection Justification:-

1. **Connectors:-**
 - **High-Current Power Terminals:** Chosen to handle high current draw reliably without overheating or causing voltage drops.
 - **Breakout & Signal Pin Headers:** 2.54 mm pitch headers provided for easy connection to microcontrollers, sensors, or auxiliary distribution channels.
2. **Fuses/Protection/Passive Modules :-**
 - **Inline Fuse Holders (F1, F2, F3):** Protects downstream electronics by breaking the circuit automatically in case of a short circuit or overcurrent spike.
 - **Thick Copper Traces:** Maximized trace width to support continuous high current delivery across the board while keeping thermal dissipation low.

~ ASSIGNMENT 4 ~

Assignment Name:- Battery Pack Specifications

Problem Statement:-

Problem 1: Go through the specifications of [this battery pack](#) and answer the questions below. Show your calculations as well.

- On average, what is the maximum continuous discharge current this battery can provide?
- What would the energy rating (in Wh) of the battery pack be if 5 cells were removed such that the configuration becomes 5S5P?
- What would the capacity (in mAh) of a 3S2P configuration of the individual cells be?
- What would the nominal voltage across the ends of this 3S2P configuration be?

GENX ULTRA 21.6V 21000MAH 20C PREMIUM LITHIUM ION RECHARGEABLE BATTERY

₹18,075.00

Current Reviews: 0 Add Your Review

GST Input Tax Credit @ 18 % of ₹3253.5 available. (Know More)

Warranty : None

Free Economy Shipping option available



Battery Pack Specifications:

Nominal voltage	21.6V
Configuration	6S5P (6cells in Series 5 cells in Parallel)
Nominal Capacity	21000mah
Continuous Discharge	20C
Internal Resistance	0.72mOhms (±)10%
Energy (Wh)	453.6
Discharge connector	AS150U
Burst Discharge	20C
Operating Temperature Range	Ambient Charge : -20 to 45°C Ambient Discharge: -40 to 60°C
Storage Temperature Range	Storage Temperature : 0 to 60°C (65% Humidity)
Cycle Life	DOD 80% : 1C/1C 800+ cycles DOD 80% : 1C/10C 500+ cycles DOD 80% : 1C/20C 300+ cycles
Standard Charge	4.2A(1C), CCCV, 4.20 V, 250mA cut-off
Rapid Charge	8.4A(2C), CCCV, 4.20 V, 100mA cut-off
Standard Discharge Current	0.5C
Dimension	228*58*80 mm (±)3% *ZigZag Cell Structure
Weight (g)	2290g (±)10g

Package Includes:

- 1 x GenX Ultra 21.6V 6S5P 21000mah 20C/20C Premium Lithium Ion Rechargeable Battery

Answers:-

Ans 1 $\Rightarrow$ The maximum continuous discharge current that this battery pack can safely provide is **420 A**.

Ans 2 $\Rightarrow$ The total energy rating of the modified 5S5P configuration drops from 453.6 Wh to **378 Wh**.

Ans 3 $\Rightarrow$ The total capacity of individual cells arranged in a 3S2P configuration is **8400 mAh**.

Ans 4 $\Rightarrow$ The nominal voltage across the terminals of this 3S2P configuration is **10.8 V**.

Calculations:-

Cell Specific Calculations $\Rightarrow$

Before answering the questions, we break down the original **6S5P** pack to find the ratings of a single individual cell:

- **Total Cells in Pack:**

$$6 \text{ Series} \times 5 \text{ Parallel} = 30 \text{ cells}$$

- **Individual Cell Nominal Voltage (V_{cell}):**

$$V_{\text{cell}} = \frac{\text{Pack Voltage}}{\text{Series Cells (6S)}} = \frac{21.6 \text{ V}}{6} = \mathbf{3.6 \text{ V}}$$

- **Individual Cell Capacity (C_{cell}):**

$$C_{\text{cell}} = \frac{\text{Pack Capacity}}{\text{Parallel Cells (5P)}} = \frac{21000 \text{ mAh}}{5} = \mathbf{4200 \text{ mAh (4.2 Ah)}}$$

- **Individual Cell Energy (E_{cell}):**

$$E_{\text{cell}} = V_{\text{cell}} \times C_{\text{cell}} = 3.6 \text{ V} \times 4.2 \text{ Ah} = \mathbf{15.12 \text{ Wh}}$$

Q1 Calculations $\Rightarrow$

1. Maximum Continuous Discharge Current

- **Formula:** Current (A) = Pack Capacity (Ah) $\times$ C-Rating
- **Calculation:**

$$21 \text{ Ah} \times 20\text{C} = \mathbf{420 \text{ A}}$$

Q2 Calculations $\Rightarrow$

2. Energy Rating of a 5S5P Configuration

- Method A (By Cell Count):
 - Total cells in a 5S5P configuration: $5 \times 5 = 25$ cells

$$\text{Total Energy} = 25 \text{ cells} \times 15.12 \text{ Wh/cell} = \mathbf{378 \text{ Wh}}$$

- Method B (By Pack Voltage & Capacity):
 - New Voltage (5S): $5 \times 3.6 \text{ V} = 18 \text{ V}$
 - New Capacity (5P): $5 \times 4.2 \text{ Ah} = 21 \text{ Ah}$

$$\text{Total Energy} = 18 \text{ V} \times 21 \text{ Ah} = \mathbf{378 \text{ Wh}}$$

Q3 Calculations $\Rightarrow$

3. Capacity of a 3S2P Configuration

- Formula: Pack Capacity = Parallel Paths (P) $\times C_{\text{cell}}$
- Calculation:

$$2P \times 4200 \text{ mAh} = \mathbf{8400 \text{ mAh (8.4 Ah)}}$$

(Note: Series configuration changes voltage, not capacity)

Q4 Calculations $\Rightarrow$

4. Nominal Voltage of a 3S2P Configuration

- Formula: Pack Voltage = Series Cells (S) $\times V_{\text{cell}}$
- Calculation:

$$3S \times 3.6 \text{ V} = 10.8 \text{ V}$$

(Note: Parallel configuration changes capacity, not voltage)

~ ASSIGNMENT 5 ~

Assignment Name:- Voltage Regulation Comparison

Problem Statement:-

Problem 2: Your primary battery pack outputs a nominal 22.2V. However, your logic systems (which include an ESP32-S3 module and some custom sensors) require a highly stable 5V supply and draw about 2 Amps of current collectively.

Explain why using a switching step-down converter (Buck Converter) is vastly superior to using a linear voltage regulator (like an LM7805 or LDO) in this specific scenario. Calculate the power that would be dissipated as heat if a linear regulator were used here.

Answer:-

- A **buck converter is superior** because it steps down voltage via energy-storage components at around 90% efficiency, whereas a **linear regulator wastes around 77.5% of the power as extreme heat**, draining the battery at a faster rate and risking component meltdown by the excess heat dissipation.
- **Calculation for Heat Dissipated (incase of Linear Regulator):-**
 - Useful load power: $P_{\text{out}} = V_{\text{out}} \times I_{\text{out}} = 5.0\text{V} \times 2.0\text{A} = 10.0\text{W}$
 - Total power drawn from battery by linear regulator: $P_{\text{in}} = V_{\text{in}} \times I_{\text{out}} = 22.2\text{V} \times 2.0\text{A} = 44.4\text{W}$
 - Power dissipated as heat: $P_{\text{dissipated}} = P_{\text{in}} - P_{\text{out}} = (22.2\text{V} - 5.0\text{V}) \times 2.0\text{A} = 17.2\text{V} \times 2.0\text{A} = \mathbf{34.4\text{W}}$
 - Linear efficiency: $\eta = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{5.0\text{V}}{22.2\text{V}} \times 100\% = \mathbf{22.5\%}$
- Therefore, a switching buck converter must be used here rather than a linear regulator, because linear regulator wastes more than triple the useful load power as pure heat (34.4W wasted vs 10W utilized), drastically reducing rover's battery runtime and introducing heating issues.

~ ASSIGNMENT 6 ~

Assignment Name:- Power Budget and Battery Selection

Problem Statement:-

Problem 3: Power Budget and Battery Selection

Part A: Power Budget

The components currently in the rover system are listed in the **Power Budget Sheet**. For each component, find and fill in the columns according to their headers.

Part B: Battery Recommendation

Based on the total power budget you have computed, recommend a battery to power all of the components simultaneously for at least **30 minutes**. Justify your recommendation by showing:

1. The minimum capacity (in Wh) required for 30 minutes of operation, with a 20% safety margin added.
2. Why you chose the specific battery (chemistry, voltage, form factor).
3. What voltage regulation stage(s) would be needed between the battery and the components.

Part A – Power Budget Table ==>

Component	Datasheet	Operational Voltage	Current Draw (cont.)	Power Draw(cont.)	Connector used
Nvidia Jetson AGX Orin	Official Spec Sheet	19 V (7 V to 20 V)	~3.16 A (at 19 V)	60 W	DC Barrel Jack / Molex
Raspberry 5 Pi	Official Spec Sheet	5 V	5.0 A	25 W	USB Type-C
ESP32-S3	Manufacturer's Datasheet	3.3 V	0.5 A	1.65 W	Micro-USB / Headers
Total Power Draw	—	—	—	86.65 W	—

Logic Behind the Component Choices:-

In this specific rover architecture, the components represent a 3-tier compute hierarchy:-

--> High-Level Compute (Nvidia Jetson AGX Orin - ~60W): Handles heavy AI, autonomous navigation, SLAM (Simultaneous Localization and Mapping), and computer vision.

--> Mid-Level Compute (Raspberry Pi 5 - ~25W): Handles high-level communication, sensor aggregation, data logging, and secondary control tasks.

--> Low-Level Microcontroller (ESP32-S3 - ~1.65W): Handles real-time motor control, safety interlocks, and reading raw sensor data with microsecond precision.

Spec Sheets vs. Datasheets:-

--> The Jetson AGX Orin and Raspberry Pi 5 reference system-level Spec Sheets because they are fully assembled single-board computers.

--> The ESP32-S3 references a component-level Datasheet because it is an individual microcontroller chip requiring low-level electrical specs.

Part B – Battery Recommendation & Justification =>

Ans 1 $\Rightarrow$ Minimum Battery Capacity Required: 52 Wh

Reasoning:-

- The rover draws 86.65 W of power total.
- Running it for 30 minutes (0.5 hours) requires 43.33 Wh of energy.
- Adding a 20% safety margin brings the total to 52 Wh.

Ans 2 $\Rightarrow$ Battery Selection & Justification: 6S Lithium-Polymer (LiPo) or Li-ion Battery (22.2 V nominal, ~3000 mAh or higher)

Reasoning:-

[Chemistry] → LiPo / Li-ion batteries offer lightweight, compact power and can safely deliver high burst currents when the rover motors or board run under full load.

[Voltage] → A 22.2 V (6S) pack provides plenty of voltage overhead to efficiently power all regulators down to 19 V, 5 V, and 3.3 V without dropping out as the battery drains.

[Form Factor] → Small and light enough to easily mount on a mobile rover chassis.

Ans 3 $\Rightarrow$ Voltage Regulation Stages Needed:-

→ 19 V Step-Down (Buck) Regulator: Converts 22.2 V from the battery down to 19 V for the Nvidia Jetson AGX Orin.

→ 5 V Step-Down (Buck) Regulator: Converts 22.2 V down to 5 V (at 5 A) for the Raspberry Pi 5.

→ 3.3 V Step-Down Regulator: Converts 5 V (or 22.2 V) down to 3.3 V for the ESP32-S3.